ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 3 SET D **WINTER SEMESTER, 2024-2025**

DURATION: 30 MINUTES **FULL MARKS: 30**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. The course teacher of Math 4341 of SWE'23 has recently observed an unusual pattern in student attendance. He has noticed that many students are frequently absent from class and is concerned that, if this trend continues, some students may not meet the attendance requirements to sit for the semester final examination. With the final exam about one month away, the instructor aims to predict students future attendance in order to identify and warn those who may be at risk.

   According to information provided by anonymous but reliable Linear Algebra enthusiasts, the **number of classes attended (r)** is influenced by the **number of hours a student spends watching reels or short-form videos (p)** and **the number of Gilbert Strang lectures the student has watched (q)**. To investigate this relationship, the following data (see Table 1) have been provided as supporting evidence, and the instructor considers this information credible.

   Table 1: Data for Attendance Prediction

Student	Reel Watch Time (p)	Class Lectures Watched (q)	Classes Attended (r)
Samiul	10	7	11
Asif	9	4	0
Emon	12	8	15
Jarif	6	3	7

   Let the equation for predicting attendance be of the form $r = mp + nq + s$, where m, n, and s are constants to be determined. Now answer the following questions:

   a) Write down the data from Table 1 in Matrix form $Ax = b$, where A is the coefficient matrix, x is the variable vector, and b is the constant vector. 1 (CO3) (PO1)

   b) Based on the data provided, determine the values of m, n, and s using the least squares regression method by solving $A\hat{x} = \hat{b}$ for $\hat{x}$. Note that original vector b may not be in the column space of A, so you need to find $\hat{x}$ for the best approximation of b, $\hat{b}$, in the column space of A. 2 (CO3) (PO1)

   c) Find out the best approximation of b, $\hat{b}$ or p, in the column space of A using the value of $\hat{x}$ obtained in the previous part. (p is the projection of b onto the column space of A ($C(A)$)) 1 (CO3) (PO1)

   d) Find out the matrix P that projects b into the column space of A ($C(A)$). 1.5 (CO3) (PO1)

   e) Write down the expressions for determining the least squares solution $\hat{x}$, projected vector p and the projection matrix P. (If you have already mentioned these in Question 1.b, Question 1.c and Question 1.d, just highlight those expressions.) 1.5 (CO3) (PO1)

   f) Find the error in your prediction, i.e. the difference in attendance between the actual data and the predicted data. 2 (CO3) (PO1)

   g) Find the Mean Square Error(MSE) $= \frac{1}{n}\sum_{i=1}^{n}(b_i - \hat{b}_i)^2$. 1 (CO3) (PO1)

2. If a student needs to attend **at least 26 classes** to be eligible to sit for the semester final examination, determine which of the following students are at risk of not meeting the attendance requirements based on your predictions from the answer obtained in question Question 1. 3 (CO2) (PO1)

Table 2: Attendance Data for Risk Assessment

Student	Reel Watch Time (p)	Class Lectures Watched (q)	Classes Attended (r)	Eligibility Status (Yes/No)
Muhaiminul	3	5		
Dhrubo	10	0		
Wasi	5	7		

3. Let X and Y be two subspaces such that $X \subset Y$ in $\mathbb{R}^n$. Let $X^\perp$ and $Y^\perp$ be the orthogonal complements of X and Y respectively in $\mathbb{R}^n$. Then is $X^\perp \subset Y^\perp$? If yes, then justify with proper reasoning supported by example, and if not, what is the correct relationship between $X^\perp$ and $Y^\perp$? 5 (CO2) (PO1)

4. Let $\vec{a}$ and $\vec{b}$ be two vectors. If $\vec{a}$ is stretched by a factor of 5 and $\vec{b}$ is shrinked by a factor of 2 (halved), then what will be its effect on particular fraction $\hat{x}$ and the Projection Matrix P. $1 + 1$ (CO2) (PO1)

5. For the provided data in Table 1, do you think instead of $r = mp + nq + s$, a better model for predicting attendance would be $r = \frac{mp}{3} + nq^3 - 5s^2$? Justify your answer mathematically. 7 (CO3) (PO1)
BONUS: 7

6. In a 23 dimensional vector space ($\mathbb{R}^{23}$), what is the maximum number of mutually orthogonal 4 dimensional subspaces in the given vector space? 3 (CO2) (PO1)
BONUS: 3

Please note that -

- **Note that: Question 1-4 are mandatory worth 20 marks and Question 5 and 6 are bonus worth additional 10 marks.**
- **Try keeping fractions instead of converting to decimals in Questions 1, 2, and 5 (except for the Projection Matrix) for more accurate results.**
- **The relationships between the names mentioned in the questions and the associated data are 'mostly' hypothetical.**